MIL-C-51280B(EA)
6 June 1977
SUPERSEDING
MIL-C-51280A(MU)
19 January 1973

MILITARY SPECIFICATION

COVER, VOICEMITTER-OUTLET VALVE, C13R1

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification describes one type of molded cover for the voicemitter-outlet valve of the M17A1 field chemical-biological mask.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

EDGEWOOD ARSENAL

E5-81-113 - Cover, Voicemitter-Outlet Valve, C13R1.
A5-2-794 - Medallion.
D136-19-14 - Puncture Tester, Natural Rubber, Q38, Assembly.
E136-19-22 - Flexibility Tester, Natural Rubber, Q37, Assembly.

PUBLICATIONS

US ARMY CHEMICAL SYSTEMS LABORATORY DIRECTIVE

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D412 - Tension Testing of Rubber.
D572 - Accelerated Aging of Vulcanized Rubber by the Oxygen-Pressure Method.
D573 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D624 - Tear Resistance of Vulcanized Rubber.

(Applications for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Preproduction. Prior to the start of regular production, a preproduction sample of covers shall be produced, using molds which have received prior Government approval, in accordance with this specification for examination and tests (see 4.3).

3.2 Materials.

3.2.1 Rubber compound. The rubber compound shall be non-blooming, except for slight wax-type bloom, and shall contain no less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.2.2 Sun-checking retardant. The rubber compound shall contain a sun-checking retardant.

3.2.3 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.2.4 Coloring materials. The rubber compound shall be colored by carbon-black, as required.

3.3 Manufacture. The cover shall be manufactured in Government approved molds so as to conform to Drawing E5-81-113.

3.3.1 Repair of flaws. Covers with small flaws may be repaired and accepted, provided there is no change in the formulation or surface continuity.

3.3.2 Test slabs. Test slabs, 6 by 6 inches, by 0.075 ± 0.01 inch, shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of covers. The slabs shall have a cure equivalent to that of the regular production covers (see 6.3).

3.4 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:05Z
Check the source to verify that this is the current version before use.

Table I. Physical properties

Property	Unaged	After Heat aging	Test paragraph
Average tensile strength, MPa, min	20.7	17.6	4.4.4.2
Average ultimate elongation, %, min	600		4.4.4.2
Percent increase (max)		20	
Percent decrease (max)		25	
Average tensile stress, MPa at 200% elongation	1.72 to 3.10	1.72 to 3.10	4.4.4.2
Individuals (min)	1.65	1.65	
Average tear resistance, kN/m, min			4.4.4.2
Die A or B	52.5	36.8	
Die C	39.4	28.9	
Average tension set at 75% ultimate elongation, %, max	20		4.4.4.2
Ozone resistance, min	80		4.4.4.3
Flexibility			
Units at 24° ± 3°C	7.5		
Units retained at -32° ± 3°C	4.8		
*Puncture resistance, psi, min	9.5		4.4.4.5

*To be performed on preproduction sample only.

3.5 Workmanship. The cover's inner and outer surfaces shall be smooth and free from imperfections, such as rips, tears, punctures, pits, bumps, porous areas, cracks, checks, flash, nicks, abrasions, or blisters. The rubber shall be compact and free from holes, bubbles, and foreign matter. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed.

3.6 Resistance to oxygen-pressure aging. The cover shall show no cracking or increased hardness or tackiness after 72 hours of accelerated aging in oxygen, at a pressure of 2.10 ± 0.10 megapascals (MPa) and a temperature of 70° ± 1°C, when tested as specified in 4.4.4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. Thy Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.2 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 25 covers, with a minimum of 2 covers from each mold cavity, and 39 test slabs shall be manufactured using the same methods, materials, equipment, and rubber formulation as will be used during regular production.

4.3.2 Inspection procedure. The sample covers shall be inspected for all requirements of the drawings and of this specification.

4.3.3 Tests. Thirty-nine test slabs, produced in accordance with 3.3.2, shall be tested in accordance with 4.4.4.1 through 4.4.4.5. One sample cover shall be tested in accordance with 4.4.4.6.

4.3.4 Mold approval. Prior to the initation of production, the supplier shall ascertain that each applicable mold has been approved by the Government.

4.3.5 Acceptance/rejection criteria. The preproduction sample shall meet the inspection and tests specified in 4.3.2 and 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:05Z
Check the source to verify that this is the currentversion before use.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the covers produced in sequence during one month, as indicated by the dots in the medallion on the cover. A sublot shall consist of the covers produced in no more than one work week from one batch of material (see 6.6). Acceptably repaired covers from any accepted parent lot may be incorporated into and assume the lot number of current production. Any major change in material, equipment, rubber formulation, or procedure used in regular production which differs from that used in the preproduction sample shall be approved by the Government prior to its institution.

4.4.2 Sampling.

4.4.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Thirty-three slabs (see 3.3.2) shall be submitted with each sublot for testing, as specified in 4.4.4.1 through 4.4.4.4. The puncture resistance test is not required. One cover shall be submitted with each sublot for testing as specified in 4.4.4.6.

4.4.3 Inspection procedure. The sample covers shall be inspected in accordance with the classification of defects and with MIL-STD-105.

4.4.3.1 Classification of defects.

(a) Cover, voicemitter-outlet valve, C13R1 (Drawing E5-81-113).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (see 3.5)
102	Rivet hole diameter and associated slot depth, including orientation and location, incorrect
103	Anti-tear hole or slit not clean or free from flash or nicks
104	Anti-tear hole or slit missing, incorrect, or incorrectly located.
Minor:	AQL 2.5 percent defective
201	Thickness incorrect

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

(b) Medallion (Drawing A5-2-794).

Categories	Defects
Critical:	None defined
Major:	None defined
Minor:	AQL 2.5 percent defective
201	Medallion information omitted or not legible

4.4.4 Tests.

4.4.4.1 Test specimens. Specimens shall not be buffed or stretched before testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for tension-set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear-resistance testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone-resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture-resistance. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 70° ± 1°C for a continuous period of 168 ± 1 hours or at a temperature of 100° ± 1°C for 24 ± 1 continuous hours, as specified in ASTM D573 (see 6.7).

4.4.4.2 Tensile strength, ultimate elongation, tensile stress, tear resistance, and tension set. Tensile strength, ultimate elongation, tensile stress, and tension set shall be determined in accordance with ASTM D412; tear-resistance shall be determined in accordance with ASTM D624. The angle-type, or cresecent type die shall be used for tear-resistance testing.

4.4.4.3 Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

4.4.4.4 Flexibility. Flexibility shall be determined using Edgewood Arsenal Q37 Natural Rubber Flexibility Tester Assembly (E136-19-22). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 24° ± 3°C and four specimens shall be tested at -32° ± 3°C after being conditioned for 72 hours at each temperature.

4.4.4.5 Puncture resistance. Resistance to hydrostatic pressure shall be determined using Edgewood Arsenal Q38 Natural Rubber Puncture Tester Assembly (D136-19-14).

4.4.4.6 Resistance to oxygen-pressure aging. The cover shall be tested in accordance with ASTM D572.

4.4.5 Acceptance/rejection criteria. If any specimen fails to meet the requirements of 3.4, a retest for the requirement failed is allowed. If the retested specimen fails, the sublot shall be rejected. If the sample fails to meet the requirement of 3.6, the sublot shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking, interplant shipment (see 6.8). The covers, voicemitter-outlet valve shall be packaged and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with FED-STD-123.

5.2 Repair parts. When covers, voicemitter-outlet valve are procured for storage and issue as repair parts, preservation, packaging, packing and marking shall be as specified on the packaging data sheet which is identified by the National Stock Number.

6. NOTES

6.1 Intended use. This specification describes a protective cover intended to be used on the M17A1 field chemical-biological mask.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Preproduction.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

1. Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
2. Name and address of test facility and shipping instructions when testing is performed by the Government.
3. Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Equivalent cure. Slabs and covers of the same rubber formulation and thickness are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than $\pm$ 2.5 percent.
(c) Their curing times can be equated in the following equation:

$$t_2 = t_1 \times 1.6^{(T_1 - T_2)/10}$$

where: t_1 = minutes of cure at temperature T_1

t_2 = minutes of cure at temperature T_2

T_1 = temperature (°F) of cure for time t_1, and

T_2 = temperature (°F) of cure for time t_2.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation of time-temperature relationships in determining equivalent cure.

6.4 Major change. The term applies to any change in formulae, processing, of curing of currently approved rubber stocks. The major change should be considered effected when one or more of the following conditions occur:

(a) Components of formulae are varied in excess of the limitations stated in the manufacturer's tolerance (see 6.5).
(b) The addition or deletion of one or more components of formulae.
(c) Substitution for one or more components of formulae without advance approval of Chemical Systems Laboratory.
(d) Deviations from the submitted and approved factory mold cure exceeding +5°F or -10°F, even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation or cure of the original samples submitted for approval.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:05Z
Check the source to verify that this is the current version before use.

EQUIVALENT CURE TIME FOR NATURAL RUBBER

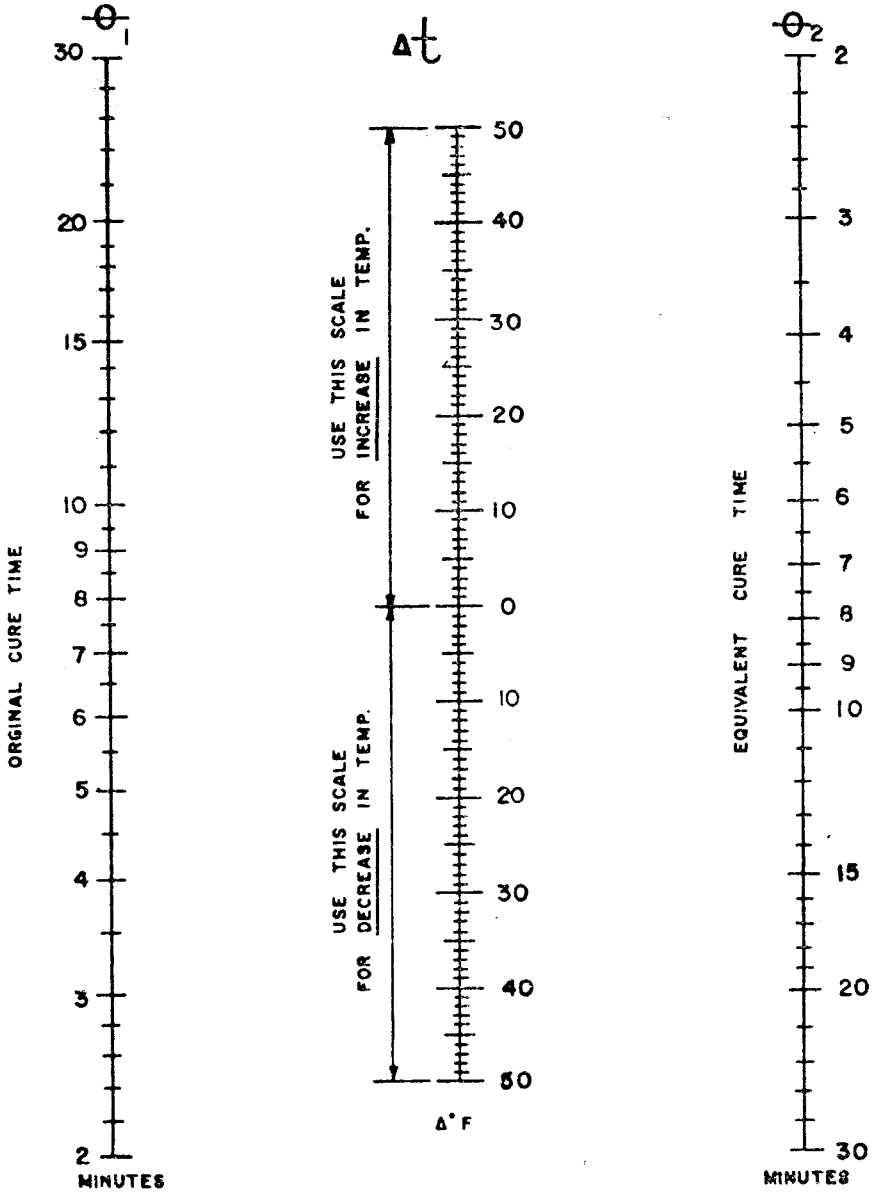


Figure 1.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS θ_2 SCALE AT THE EQUIVALENT CURE TIME.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:05Z
Check the source to verify that this is the current version before use.

6.5 Manufacturer's tolerances. The manufacturer should submit to Chemical Systems Laboratory the variations in amounts used, or tolerances, for accelarators, plasticizers, and softeners. These tolerances are permitted to enable the manufacturer to produce stock which will consistently comply with this specification and which will process satisfactorily in the plant. Components of formulae may be submitted on the basis of percentage of stock or number of parts of rubber, and the manufacturer's tolerances recorded in the same units.

6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation to make the final product substantially uniform.

6.7 Aging. Aging at 100° ± 1°C for a continuous period of 24 hours is slightly more severe on rubber than aging at 70° ± 1°C for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 70° ± 1°C for 168 consecutive hours.

6.8 Interplant shipments. Packaging for supplies and materials which will not enter the military supply system. Typical interplant shipments are shipments from the vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A700

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:05Z
Check the source to verify that this is the current version before use.